Practical 13: PHP JSON API and AJAX/Fetch-Based CRUD without Page Reload

Problem Definition

Create simple PHP JSON API endpoints for events or students and consume them using Fetch API. Perform list, add, update, and delete operations asynchronously without full page reload.

Key questions
1. Do PHP endpoints return valid JSON responses?
PHP endpoints return appropriate responses for event operations.

2. Are GET and POST requests handled properly?
GET and POST requests are handled for CRUD operations.

3. Are loading, success, and error states shown on the frontend?
Success and error messages are displayed on the frontend.

4. Is input validation performed before database operations?
Yes, input validation is performed before database operations.

Code
Db.php

```php
<?php
$conn = mysqli_connect(
    "localhost",
    "root",
    "",
    "event_api_db"
);

if (!$conn) {
    die("Database connection failed");
}

mysqli_set_charset($conn, "utf8mb4");
?>
```

Api.php

```php
<?php
header("Content-Type: application/json; charset=UTF-8");

include "db.php";

$method = $_SERVER["REQUEST_METHOD"];

if ($method == "GET") {

    $result = mysqli_query(
        $conn,
        "SELECT * FROM events ORDER BY id DESC"
    );

    $events = [];

    while ($row = mysqli_fetch_assoc($result)) {
        $events[] = $row;
    }

    echo json_encode($events);
    exit;
}

if ($method == "POST") {

    $data = json_decode(file_get_contents("php://input"), true);

    if (!is_array($data) || !isset($data["action"])) {
        http_response_code(400);
        echo json_encode([
            "success" => false,
            "message" => "Invalid request"
        ]);
        exit;
    }

    $action = $data["action"];
```

```php
if ($action == "add") {

    $name = trim($data["event_name"] ?? "");
    $date = $data["event_date"] ?? "";

    if ($name == "" || !preg_match("/^\d{4}-\d{2}-\d{2}$/", $date)
        || !checkdate((int)substr($date, 5, 2),
                      (int)substr($date, 8, 2),
                      (int)substr($date, 0, 4))) {
        http_response_code(400);
        echo json_encode([
            "success" => false,
            "message" => "Enter valid event name and date"
        ]);
        exit;
    }

    $stmt = mysqli_prepare(
        $conn,
        "INSERT INTO events (event_name, event_date) VALUES (?, ?)"
    );

    mysqli_stmt_bind_param($stmt, "ss", $name, $date);
    $ok = mysqli_stmt_execute($stmt);

    echo json_encode([
        "success" => $ok,
        "message" => $ok ? "Event added successfully" : "Add failed"
    ]);
    exit;
}

if ($action == "update") {

    $id = filter_var($data["id"] ?? null, FILTER_VALIDATE_INT);
    $name = trim($data["event_name"] ?? "");
    $date = $data["event_date"] ?? "";

    if (!$id || $name == "" ||
        !preg_match("/^\d{4}-\d{2}-\d{2}$/", $date) ||
```

```
            !checkdate((int)substr($date, 5, 2),
                       (int)substr($date, 8, 2),
                       (int)substr($date, 0, 4))) {
            http_response_code(400);
            echo json_encode([
                "success" => false,
                "message" => "Invalid event details"
            ]);
            exit;
        }

        $stmt = mysqli_prepare(
            $conn,
            "UPDATE events SET event_name=?, event_date=? WHERE id=?"
        );

        mysqli_stmt_bind_param($stmt, "ssi", $name, $date, $id);
        $ok = mysqli_stmt_execute($stmt);

        echo json_encode([
            "success" => $ok && mysqli_stmt_affected_rows($stmt) >= 0,
            "message" => $ok ? "Event updated successfully" : "Update
failed"
        ]);
        exit;
    }

    if ($action == "delete") {

        $id = filter_var($data["id"] ?? null, FILTER_VALIDATE_INT);

        if (!$id) {
            http_response_code(400);
            echo json_encode([
                "success" => false,
                "message" => "Invalid event ID"
            ]);
            exit;
        }
```

```
        $stmt = mysqli_prepare(
            $conn,
            "DELETE FROM events WHERE id=?"
        );

        mysqli_stmt_bind_param($stmt, "i", $id);
        $ok = mysqli_stmt_execute($stmt);

        echo json_encode([
            "success" => $ok && mysqli_stmt_affected_rows($stmt) > 0,
            "message" => $ok ? "Event deleted successfully" : "Delete
failed"
        ]);
        exit;
    }

    http_response_code(400);
    echo json_encode([
        "success" => false,
        "message" => "Unknown action"
    ]);
    exit;
}

http_response_code(405);
echo json_encode([
    "success" => false,
    "message" => "Method not allowed"
]);
?>
```

Index.html

```
```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Event Management API</title>
```
```

```
<style>
    body {
        font-family: Arial, sans-serif;
        margin: 30px;
        background: #f2f2f2;
    }

    .box {
        max-width: 700px;
        margin: auto;
        background: white;
        padding: 20px;
        border-radius: 10px;
    }

    input, button {
        padding: 10px;
        margin: 5px 0;
    }

    input {
        width: 95%;
    }

    button {
        cursor: pointer;
    }

    table {
        width: 100%;
        margin-top: 20px;
        border-collapse: collapse;
    }

    th, td {
        padding: 10px;
        border: 1px solid #ccc;
        text-align: left;
    }
```

```
        #message {
            margin-top: 10px;
        }
    </style>
</head>

<body>

<div class="box">

    <h2>Event Management System</h2>

    <form id="eventForm">

        <input type="hidden" id="eventId">

        <input
            type="text"
            id="eventName"
            placeholder="Enter Event Name"
            required
        >

        <input
            type="date"
            id="eventDate"
            required
        >

        <button type="submit" id="saveButton">
            Add Event
        </button>

        <button type="button" id="cancelButton" hidden>
            Cancel Edit
        </button>

    </form>
```

```
    <p id="message" role="status"></p>

    <h3>Event List</h3>

    <button type="button" onclick="loadEvents()">
        Refresh List
    </button>

    <table>
        <thead>
            <tr>
                <th>ID</th>
                <th>Event Name</th>
                <th>Date</th>
                <th>Actions</th>
            </tr>
        </thead>

        <tbody id="eventList"></tbody>
    </table>

</div>

<script>
const API = "api.php";

const form = document.getElementById("eventForm");
const eventId = document.getElementById("eventId");
const eventName = document.getElementById("eventName");
const eventDate = document.getElementById("eventDate");
const eventList = document.getElementById("eventList");
const message = document.getElementById("message");
const saveButton = document.getElementById("saveButton");
const cancelButton = document.getElementById("cancelButton");

function showMessage(text) {
    message.textContent = text;
}

async function requestAPI(method, data = null) {
```

```
    const options = { method };

    if (data !== null) {
        options.headers = {
            "Content-Type": "application/json"
        };

        options.body = JSON.stringify(data);
    }

    const response = await fetch(API, options);
    const result = await response.json();

    if (!response.ok) {
        throw new Error(result.message || "Request failed");
    }

    return result;
}

// LIST
async function loadEvents() {
    showMessage("Loading events...");

    try {
        const events = await requestAPI("GET");

        eventList.replaceChildren();

        events.forEach(event => {
            const tr = document.createElement("tr");

            const idCell = document.createElement("td");
            idCell.textContent = event.id;

            const nameCell = document.createElement("td");
            nameCell.textContent = event.event_name;

            const dateCell = document.createElement("td");
            dateCell.textContent = event.event_date;
```

```
            const actionCell = document.createElement("td");

            const editButton = document.createElement("button");
            editButton.textContent = "Edit";

            editButton.onclick = () => editEvent(event);

            const deleteButton = document.createElement("button");
            deleteButton.textContent = "Delete";

            deleteButton.onclick = () => deleteEvent(event.id);

            actionCell.append(editButton, deleteButton);
            tr.append(idCell, nameCell, dateCell, actionCell);

            eventList.appendChild(tr);
        });

        showMessage("Events loaded successfully");

    } catch (error) {
        showMessage(error.message);
    }
}

// ADD AND UPDATE
form.addEventListener("submit", async function(e) {
    e.preventDefault();

    const id = eventId.value;

    const data = {
        action: id ? "update" : "add",
        event_name: eventName.value.trim(),
        event_date: eventDate.value
    };

    if (id) {
        data.id = Number(id);
```

```
    }

    if (!data.event_name || !data.event_date) {
        showMessage("Please fill all fields");
        return;
    }

    saveButton.disabled = true;
    showMessage("Saving event...");

    try {
        const result = await requestAPI("POST", data);

        showMessage(result.message);

        form.reset();
        resetForm();

        await loadEvents();

    } catch (error) {
        showMessage(error.message);

    } finally {
        saveButton.disabled = false;
    }
});

// LOAD DETAILS INTO FORM
function editEvent(event) {
    eventId.value = event.id;
    eventName.value = event.event_name;
    eventDate.value = event.event_date;

    saveButton.textContent = "Update Event";
    cancelButton.hidden = false;

    showMessage("Edit event details and click Update");
}
```

```
// CANCEL EDIT
function resetForm() {
    eventId.value = "";
    saveButton.textContent = "Add Event";
    cancelButton.hidden = true;
}

cancelButton.addEventListener("click", function() {
    form.reset();
    resetForm();
    showMessage("Edit cancelled");
});

// DELETE
async function deleteEvent(id) {
    if (!confirm("Are you sure you want to delete this event?")) {
        return;
    }

    showMessage("Deleting event...");

    try {
        const result = await requestAPI("POST", {
            action: "delete",
            id: Number(id)
        });

        showMessage(result.message);
        await loadEvents();

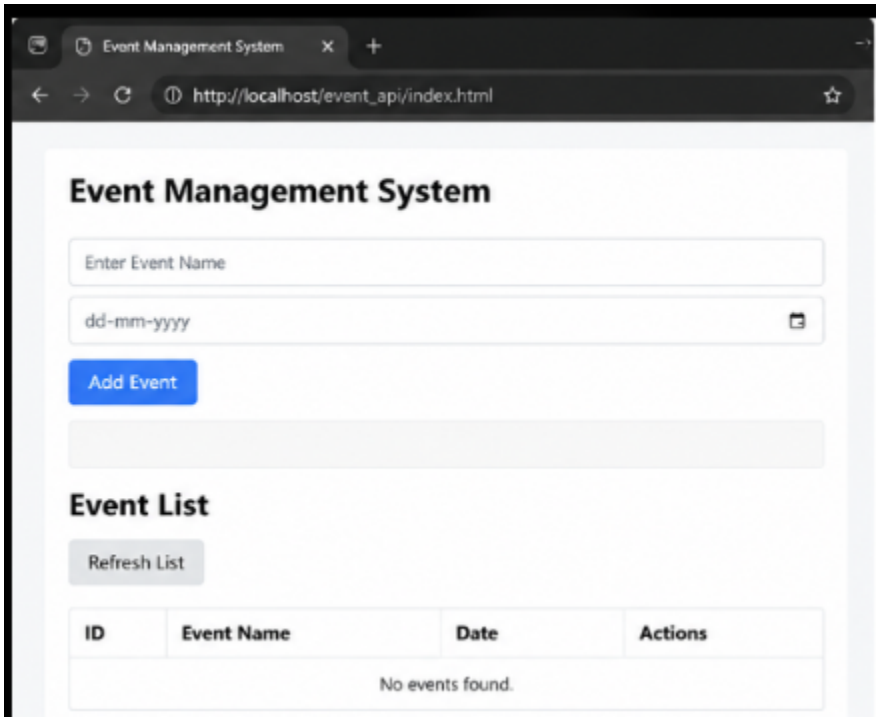
    } catch (error) {
        showMessage(error.message);
    }
}

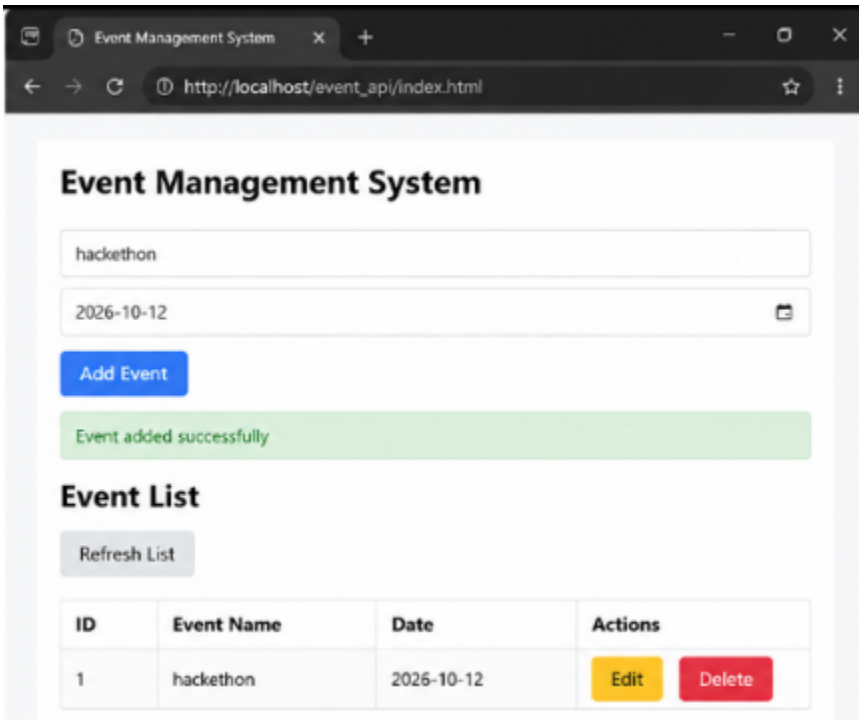
// INITIAL LOAD
loadEvents();
</script>

</body>
```

```
</html>
```

Output


Event Management System
http://localhost/event_api/index.html
Event Management System
Enter Event Name
dd-mm-yyyy
Add Event
Event List
Refresh List
ID
Event Name
Date
Actions
No events found.


Event Management System
http://localhost/event_api/index.html
Event Management System
hackethon
2026-10-12
Add Event
Event added successfully
Event List
Refresh List
ID
Event Name
Date
Actions
1
hackethon
2026-10-12
Edit
Delete

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